

# Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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## Abstract

Linear probes show emotion is decodable in the middle-to-late layers of frozen speech transformers, but not which internal features carry it, nor whether the signal is acoustic or lexical. We train TopK Sparse Autoencoders (SAEs) on *per-frame* activations of six frozen large encoders and dissect their emotion features on the lexicon-controlled CREMA-D corpus (91 actors  $\times$  12 fixed sentences  $\times$  6 emotions). A confound-controlled selection—a feature’s firing must be more informative about emotion than about the spoken sentence or the speaker—isolates genuine emotion features from phonetic and speaker-driven ones. We find that (i) emotion *disentangles* into clean monosemantic features earlier (L2–L7) than where it is best linearly *decoded* (L10); (ii) all six encoders share an acoustic emotion-feature basis that generalizes across unseen speakers; and (iii) the recovered emotion signal is acoustic, not lexical. On the EMIS incongruent-speech benchmark we further localize the lexical shortcut to depth: it emerges only in the deep layers of the ASR-trained encoders (HuBERT/WavLM/wav2vec2), while Whisper stays audio-grounded at every layer. Decodability and disentanglement are thus distinct, and emotion here is carried by speaker-invariant acoustic features.

## Keywords

speech emotion recognition, sparse autoencoders, mechanistic interpretability, probing, acoustic features, disentanglement

## 1 Introduction

Self-supervised speech encoders dominate speech emotion recognition (SER), yet were trained on ASR-style objectives that encode substantial linguistic structure. Layer-wise probing answers *where* emotion is readable, but cannot say *which* features encode it, whether the depth pattern holds across encoders, speakers and languages, or whether the signal a probe exploits is acoustic (how something is said) or lexical (what is said). Sparse autoencoders (SAEs) decompose dense activations into an over-complete dictionary of sparse, near-monosemantic features [2, 5], turning these questions from decodability into mechanism. Because SAEs trained on generic speech tend to rediscover *phonemes* rather than affect [1], we anchor the study on CREMA-D [3], whose 12 fixed carrier sentences hold text constant across emotions: any emotion signal recovered there is necessarily acoustic, giving a built-in positive control. We contribute (1) a confound-controlled emotion-feature definition for speech SAEs, and a three-part answer showing that (2) disentanglement and decodability peak at different depths, and (3) the emotion features are acoustic and speaker-invariant.

## 2 Method

**Per-frame activations.** The cached embeddings are mean-pooled per utterance—too few to fit an  $8\times$  over-complete SAE—so we re-extract *un-pooled* per-frame hidden states from the six frozen

large encoders (HuBERT [6], WavLM, wav2vec2, w2v-BERT, MERT, Whisper [8]): 25 states per frame (CNN L0 + 24 transformer layers), Whisper padding masked. CREMA-D layer 13 alone gives 940,273 frames. We train a TopK SAE ( $d_{in}=1024$ ,  $d_{sae}=8192$ ,  $k=32$ ) with unit-norm decoder, input standardisation, and an AuxK loss reviving dead features; it reaches FVU = 0.070 (93% variance explained) with 4/8192 dead features.

**Confound-controlled emotion features.** We score each feature by the mutual information (MI) between its firing and the emotion label, thresholded at  $5\sigma$  over a label-permutation null. Significance plus monosemanticity ( $\geq 0.5$  mass on one emotion) flags 394 features—but ~66% carry *more* information about the spoken sentence (phonetic content) or the speaker than about emotion. Requiring  $MI(\text{emotion}) > MI(\text{sentence})$  and  $> MI(\text{actor})$  removes them, leaving 61 genuine emotion features (median monosemanticity 0.59), dominated by anger and sadness; fear and neutral fall to zero, consistent with acoustic encoding of high-arousal emotions. For each, we compare the cross-validated ROC-AUC of a logistic classifier predicting its firing from eGeMAPS descriptors [4] (acoustic) vs. carrier-sentence identity (lexical);  $R^2$  is unusable as the features are ~96% zero.

## 3 Results

### 3.1 Depth: decode $\neq$ disentangle (Q i)

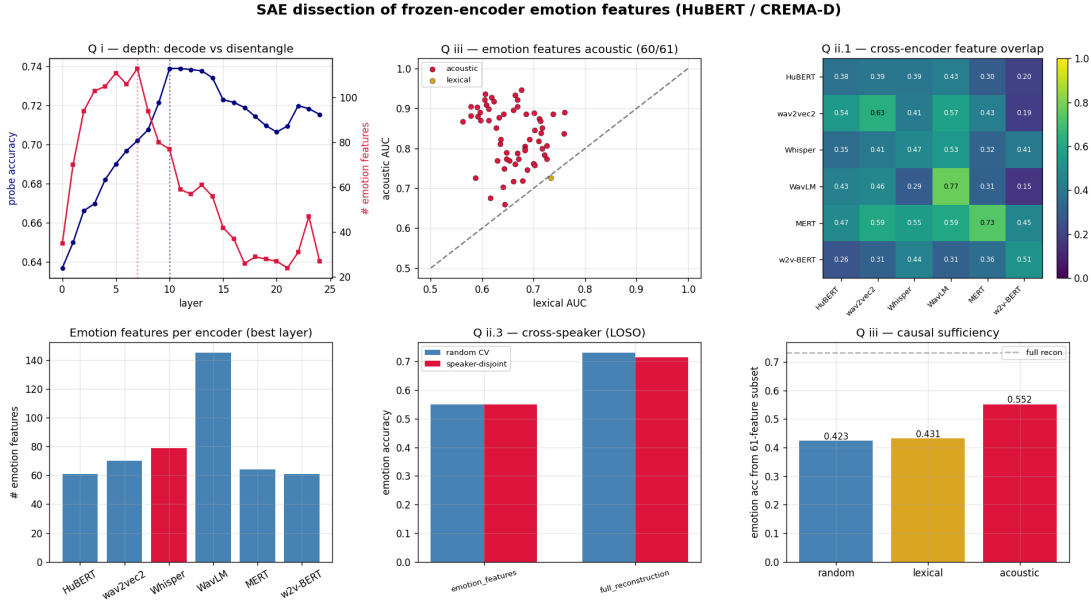
A 25-layer sweep separates two curves (Fig. 1, top-left; Table 1). Probe accuracy peaks at L10 and stays high into the deepest layers, but the *count* of monosemantic emotion features peaks earlier at L7 (and monosemanticity at L2). The deep layers L17–24 retain decodability (~0.72) yet collapse to 24–47 emotion features: emotion stays linearly readable while becoming distributed rather than packed into clean directions. *It organises into interpretable features before the depth at which a probe reads it best.*

### 3.2 Across paradigms and speakers (Q ii)

All six encoders carry emotion features at their best layer (Fig. 1, bottom-left), each anger/sadness-dominated. Matching features by their per-utterance firing correlation, the Whisper→SSL overlap (0.40) equals the SSL→SSL overlap (0.39): on fixed-lexicon CREMA-D Whisper is *not* distinct, so its known divergence concerns the lexical shortcut, not acoustic emotion encoding. Under speaker-disjoint cross-validation across the 91 actors emotion accuracy is unchanged (leave-speaker-out drop  $\approx 0$ ; ~95% actor coverage); confound control already removed the speaker-entangled features behind the published MESD leave-one-speaker-out collapse (~0.30).

### 3.3 Acoustic, not lexical (Q iii)

The positive control (fixed lexicon  $\Rightarrow$  acoustic) labels 60/61 emotion features acoustic (eGeMAPS firing-AUC 0.83 vs. sentence 0.66). Causally (Fig. 1, bottom-right), emotion decodes from the 61 acoustic features at 0.552, vs. 0.431 from lexical and 0.423 from random



**Figure 1: SAE dissection of frozen-encoder emotion features (HuBERT / CREMA-D).** Top, left to right: depth decode-vs-disentangle (probe accuracy vs. # monosemantic emotion features per layer); emotion features acoustic vs. lexical (eGeMAPS vs. sentence firing-AUC, points above the diagonal are acoustic); cross-encoder feature-overlap matrix. Bottom: emotion features per encoder at its best layer; cross-speaker (leave-speaker-out) generalization vs. random CV; causal sufficiency of acoustic vs. lexical vs. random feature subsets.

features (6-class; full reconstruction 0.73). Since the signal is distributed, sufficiency rather than ablation is the informative test: the recovered emotion signal is acoustic.

### 3.4 EMIS: the lexical shortcut is a deep-layer phenomenon (Q iv)

On EMIS (1,248 clips with text and prosodic emotion set independently) we train an emotion probe on congruent clips and test on incongruent ones; text-bias = text accuracy minus audio accuracy (positive lexical shortcut). On raw mean-pooled activations every encoder is audio-grounded in its early/mid layers; the shortcut emerges only in the DEEP layers of the ASR-trained SSL encoders (HuBERT L20 +0.79, WavLM L20 +0.73, wav2vec2 +0.87), whereas Whisper stays audio-grounded at every depth (L24 0.11) and the music model MERT and w2v-BERT never take it (Fig. 2). This reproduces the published per-encoder text-bias (HuBERT +0.72 ... Whisper 0.19) and localizes it to depth. A SAE-reconstruction probe spuriously inverted Whisper, so we report raw activations.

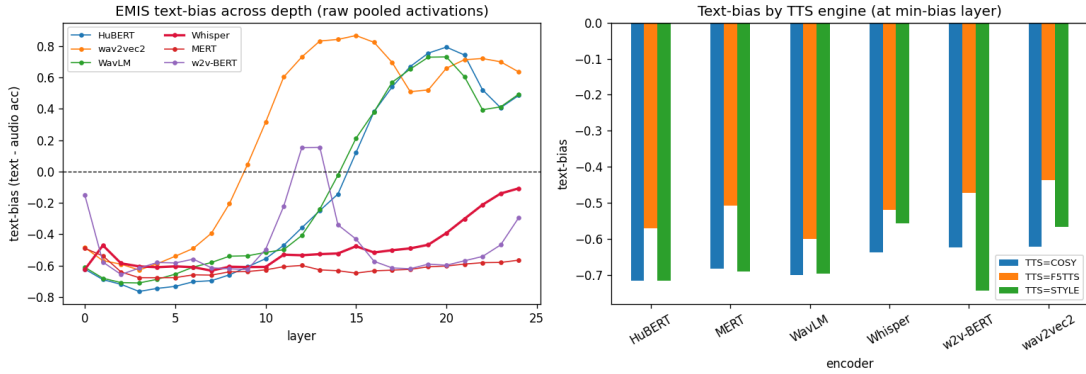
## 4 Discussion and Limitations

Decodability and disentanglement are distinct axes: a probe reads emotion best where the representation is linearly separable (L10 and the deep layers), but the encoder packs emotion into clean monosemantic features earlier (L2–L7), sharpening the prior “middle-to-late” decodability result into a claim about where emotion is organised. On lexicon-controlled data all six encoders converge on a shared acoustic basis whose features survive speaker-disjoint

evaluation; the Whisper-vs-SSL divergence concerns the lexical shortcut [7], which CREMA-D cannot pose. The EMIS experiment (§ Q iv) does pose it and confirms the prior text-bias result, additionally localizing the shortcut to the deep layers of the ASR-trained encoders — a depth that Whisper never reaches. A methodological caution: probing the SAE *reconstruction* spuriously inverted Whisper’s EMIS text-bias, so the layer sweep is run on raw activations; the SAE is the right tool for the per-feature CREMA-D analyses but not as a stand-in for raw activations in this probe. Two experiments await cluster data, each with a runnable stub: the IEMOCAP real-transcript ablation and the SpeechCraft cross-lingual arm. All

**Table 1: Key quantities (HuBERT / CREMA-D, layer 13 unless noted).**

| Quantity                                     | Value             |
|----------------------------------------------|-------------------|
| SAE reconstruction FVU (variance expl.)      | 0.070 (93%)       |
| SAE size / sparsity / dead features          | 8192 / $k=32$ / 4 |
| Emotion feats: sig → mono → confound         | 6407 → 394 → 61   |
| Decode peak / disentangle peak / mono peak   | L10 / L7 / L2     |
| Cross-encoder overlap: Whisper→SSL / SSL→SSL | 0.40 / 0.39       |
| Acoustic features (positive control)         | 60/61             |
| mean firing-AUC acoustic / lexical           | 0.83 / 0.66       |
| Sufficiency: acoustic / lexical / random     | 0.552/0.431/0.423 |
| Cross-speaker LOSO drop / actor coverage     | ≈ 0 / 0.95        |
| EMIS text-bias: Whisper L24 / ASR-SSL        | −0.11 / +0.80     |



**Figure 2: EMIS text-bias (text minus audio accuracy; train congruent / test incongruent). Left: across all 25 layers — the lexical shortcut emerges only in the deep layers of HuBERT/WavLM/wav2vec2, while Whisper (red) stays audio-grounded throughout and MERT/w2v-BERT never take it. Right: text-bias by TTS engine at each encoder’s most audio-grounded layer.**

artifacts—activations, SAEs, per-step CSVs/plots/metrics, and this paper—are released for reproduction.

## References

- [1] Anonymous. 2026. Sparse Autoencoders for Interpreting Whisper ASR Representations. arXiv:2605.12225.
- [2] Joseph Bloom, David Chanin, et al. 2024. SAELens: Training and Analyzing Sparse Autoencoders. <https://github.com/decoderresearch/SAELens>.
- [3] Houwei Cao, David G Cooper, Michael K Keutmann, Ruben C Gur, Ani Nenkova, and Ragini Verma. 2014. CREMA-D: Crowd-sourced emotional multimodal actors dataset. *IEEE Transactions on Affective Computing* 5, 4 (2014), 377–390.
- [4] Florian Eyben, Klaus R Scherer, Björn W Schuller, et al. 2016. The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for voice research and affective computing. *IEEE Transactions on Affective Computing* 7, 2 (2016), 190–202.
- [5] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2025. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.
- [6] Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021), 3451–3460.
- [7] et al. Ma. 2026. Mechanistic Interpretability of LoRA-adapted Whisper for Speech Emotion Recognition. arXiv:2509.08454.
- [8] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*.